Parkinson's Disease Detection Using Deep Learning Vision Module with Transformer

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***Abstract*—Millions of people worldwide suffer from Parkinson's disease (PD), a degenerative neurological condition. Timely intervention and better patient outcomes depend on early identification. However, subjectivity and latency are common problems with conventional diagnostic techniques, such as clinical exams and neuroimaging evaluations. This article presents an automated PD detection approach based on deep learning. We use transfer learning to three pre-trained convolutional neural networks (CNNs) in order to classify MRI images and VGG16, DenseNet121, and EfficientNetB0. DenseNet121 incorporates a Transformer-based Multi-Head Attention block to improve feature extraction. According to experimental data, the hybrid DenseNet121 + Transformer model outperforms baseline models with a validation accuracy of 96.99%. Our results demonstrate how CNNs and attention processes may be used to diagnose Parkinson's disease accurately and early, enabling AI-driven medical treatments.**

***Index Terms*—Parkinson's Disease, Deep Learning, Transfer Learning, Transformer, DenseNet121, Medical Imaging**

I. Introduction

The severe neurological disorder known as Parkinson's disease (PD) mostly affects motor functions, such as bradykinesia, stiffness, and tremors [1], [2]. Degeneration of dopamine-producing neurons in the midbrain's substantia nigra is the cause of this motor dysfunction. In addition to motor symptoms, Parkinson's disease (PD) can include non-motor symptoms such anxiety, depression, sleep disturbances, and cognitive impairment, all of which have a substantial negative impact on patients' quality of life [1]. Over 10 million people worldwide are thought to have Parkinson's disease (PD), and as the world's population ages, its prevalence is predicted to rise [2].

Early and accurate diagnosis is critical for slowing disease progression and effectively managing symptoms. While there is currently no cure for PD, timely intervention with medication, lifestyle adjustments, and supportive therapies can alleviate symptoms and enhance patients' well-being. However, conventional diagnostic methods, including clinical assessments and imaging techniques like PET and DaTScan, are expensive, invasive, and often detect the disease at advanced stages, when substantial neuronal damage has already occurred. Clinical diagnosis heavily relies on observing motor symptoms, which may not appear until a significant portion of dopamine-producing neurons has been lost [1].

Advances in artificial intelligence (AI) and deep learning offer promising avenues for automating disease detection across various medical domains [8], [10]. Convolutional neural networks (CNNs), in particular, have demonstrated exceptional capabilities in extracting intricate patterns from medical imaging data, such as magnetic resonance imaging (MRI) [8], [9]. MRI provides detailed anatomical information about the brain and can reveal subtle structural changes associated with PD [9]. This study explores the integration of CNN models with Transformer-based attention mechanisms [6] to enhance PD detection performance from MRI scans, building upon recent work in this area [11]. By combining the strengths of CNNs and Transformers, we aim to develop a more accurate and robust system for early PD detection.

II. Related Work

Deep learning, particularly Convolutional Neural Networks (CNNs), is widely used in Parkinson's Disease (PD) detection, especially with imaging data, due to their strength in extracting rich spatial features [8], [10], [11]. Various deep learning architectures have been employed for PD detection from imaging data, including combinations of CNNs and LSTMs, CNN-RNNs, and 3D CNNs, showing promising results [10], [11]. These approaches align with broader trends in deep learning applications for medical imaging, as discussed in comprehensive reviews [8]. The application of neuroimaging techniques, including MRI, for PD diagnosis is an active area of research [9].

Beyond imaging, voice and audio-based analysis has also been explored as PD often affects speech in its initial stages. Techniques like BLSTM models have been utilized to capture dynamic speech features for PD detection [10]. Analysis of handwriting and spiral drawings, which are affected by

PD's impact on fine motor skills, also utilizes deep learning models such as CNNs and RNNs for feature extraction and diagnosis [10]. Furthermore, lightweight deep learning models are being developed for potential real-time and mobile screening applications, often incorporating optimization and attention mechanisms to improve efficiency and accuracy [10]. These technological advancements are grounded in extensive research on PD biomarkers, genetics, and long-term treatment strategies.

III. Proposed Methodology

The proposed methodology uses deep learning to classify MRI scans as either Parkinson's disease or healthy. The workflow includes data preprocessing, model architecture selection, training, and evaluation.

The Parkinson's dataset utilized in our project is a classified dataset consisting of approximately 221 images of affected individuals and 610 images of normal (unaffected) individuals. Due to this significant imbalance between the two classes, it was important to address the disparity to prevent the model from becoming biased towards the majority class. To achieve a more balanced dataset and improve the model's ability to generalize, we employed various data augmentation techniques. Specifically, we applied operations such as resizing the images to a consistent dimension and normalization to standardize the pixel intensity values. These augmentation strategies not only helped in balancing the dataset but also enhanced the overall diversity and quality of the training data, leading to better model performance.

A. Dataset and Preprocessing

The MRI dataset consists of brain images categorized into Parkinson's disease and healthy classes. These images were acquired using standard MRI protocols and represent structural information about the brain [9]. Preprocessing is a crucial step to ensure that the images are in a suitable format for training the deep learning models and to improve the quality of the data [8].

1) MRI Image Acquisition: MRI scans provide detailed 3D views of the brain. T1-weighted MRI is particularly useful for examining brain anatomy [9].

2) Data Preprocessing and Augmentation:: A number of preprocessing and augmentation methods were used to get the dataset ready for training. To ensure compatibility with the CNN architectures utilized in this work, all photos were first shrunk to a standard resolution of 224×224 pixels in order to preserve uniformity across the dataset. In order to promote faster convergence during training and avoid huge pixel values controlling the learning process, pixel intensities were then normalized to the range [0,1]. The normalized picture, I_{norm}, is produced by modifying the pixel values in accordance with the original image, I. Data augmentation methods including rotation, random flipping, and brightness modifications were used to further increase the model's resilience.By artificially raising the training set's variability, these augmentations lessened overfitting and enhanced the model's capacity to generalize to new data.If I is the original image, the normalized image I_{norm} is calculated as:

$$I_{norm} = \frac{I - \min(I)}{\max(I) - \min(I)}$$

B. Model Architectures

We evaluated three pre-trained convolutional neural network (CNN) architectures for the task of Parkinson's disease detection:

- **VGG16**: A deep convolutional neural network with a simple and regular architecture, consisting of stacked convolutional layers with small 3x3 filters [3]. VGG16 is known for its ability to learn hierarchical representations of images but has a large number of parameters, which can make it computationally expensive and prone to overfitting.
- **DenseNet121**: The addition of intricate connections between a convolutional neural network's layers [4]. In a dense network, every layer receives input from every layer that came before it and sends its output to every level that follows. It's a dense connection reduces the vanishing gradient problem, promotes feature reuse, and improves information flow throughout the network. DenseNet121 is widely recognized for its efficiency and ability to extract complex patterns from relatively little samples.
- **EfficientNetB0**:Compound scaling is used in the architecture of this lightweight convolutional neural network [5]. EfficientNetB0 improves accuracy and efficiency by balancedly optimizing the network's depth, width, and resolution. Compared to existing CNN architectures, EfficientNet models have produced state-of-the-art results on a variety of image classification tasks with fewer parameters and less computational expense.

C. DenseNet121 with Transformer-Based Attention

The Transformer architecture, which was first proposed for natural language processing tasks, has shown great promise in computer vision as well [6], [7]. The attention mechanism is a crucial component of a Transformer, allowing the model to selectively focus on the most relevant parts of the input [6]. To further improve the performance of the DenseNet121 model, we added a Transformer-based Multi-Head Attention block [11].

We incorporated a Multi-Head Self-Attention block into the DenseNet121 design for this investigation. In order to capture spatial linkages and dependencies between various areas of the MRI image, the model is able to attend to different sections of the feature maps that were recovered by the DenseNet121 network thanks to the Multi-Head Attention method [6]. This allows the model to reduce noisy or unnecessary input while concentrating on the most noticeable characteristics that are suggestive of Parkinson's disease.

The architecture of the upgraded DenseNet121 model with the Transformer-based Multi-Head Attention block is shown in Figure 1. The Transformer block, which has many attention

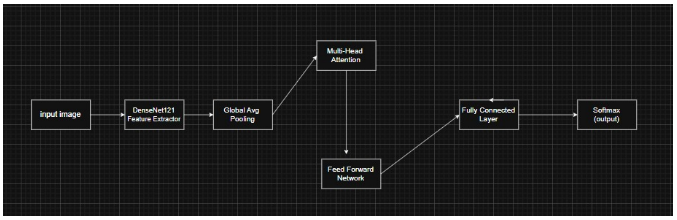

Fig. 1. Architecture of DenseNet121 with Transformer-Based multi head attention layer

heads, receives the output from the DenseNet121 network. Each attention head computes the weighted sum of the input features using weights based on how similar the query, key, and value vectors are. The outputs of the several attention heads are then concatenated and sent through a linear layer to get the final output. Classification is then done using this output. The addition of the Transformer block allows the model to capture both local and global features in the MRI images. The CNN layers in DenseNet121 extract local features, such as edges, textures, and shapes, while the Transformer block captures long-range dependencies and spatial relationships between different regions of the image. This combination of local and global feature extraction leads to a more robust and accurate representation of the MRI data, which in turn improves the performance of the Parkinson's disease detection system.

IV. Experimental Results and Analysis

We conducted many experiments to evaluate the performance of the proposed approach on the MRI dataset. The dataset was divided into training and validation sets, with 80% of the dataset being used for training and 20% for validation. The training set was used to train deep learning models, while the validation set was used to monitor the models' performance during training and modify hyperparameters.

The models were trained using the Adam optimizer, a popular optimization method for deep neural network training. The learning rate was initially set at 0.0001 and was lowered throughout training using a learning rate scheduler. For training, the binary cross-entropy loss—a well-liked loss function for binary classification problems—was used.

The models were trained for a maximum of 50 epochs before being terminated early to prevent overfitting. When the model's performance on the validation set starts to decline, early stopping is a technique that stops training. This reduces the likelihood that the model will memorize the training set and improves its ability to generalize to new data.

All experiments were conducted using Google Colab, a cloud-based platform that provides free GPUs. NVIDIA Tesla T4 GPUs were used to train the deep learning models. The significant speedup offered by the use of GPUs allowed us to train the models in a reasonable amount of time.

The specific experimental setup is summarized below:

- **Data Split**: 80% training, 20% validation.
- **Batch Size**: 32

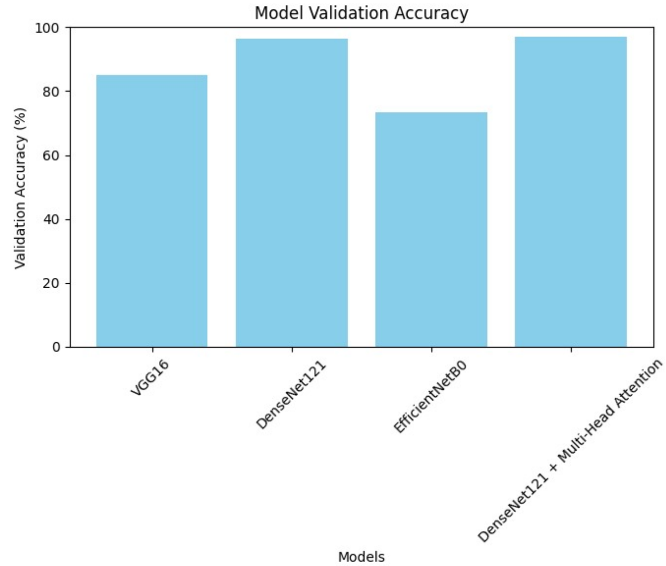

Fig. 2. Model Accuracy Comparison

- **Optimizer**: Adam
- **Learning Rate**: 0.0001 with scheduler
- **Loss Function**: Binary Cross-Entropy
- **Epochs**: 50 with Early Stopping

A. Model Performance

The validation set was used to assess the various models' performance. Each model's validation accuracy, or the proportion of properly identified MRI images, was assessed. One popular statistic for assessing how well categorization models function is validation accuracy.

Table I shows the validation accuracy of the four models: VGG16, DenseNet121, EfficientNetB0, and DenseNet121 + Transformer. As can be seen from the table, the DenseNet121 + Transformer model achieved the highest validation accuracy of 96.99%, outperforming the other three models.

TABLE I
Model Validation Accuracy Comparison

Model	Validation Accuracy
VGG16	84.94%
DenseNet121	96.39%
EfficientNetB0	73.49%
DenseNet121 + Transformer	**96.99%**

Figure 2 provides a visual comparison of the models' performance.

The VGG16 model [3] achieved a validation accuracy of 84.94%. While it demonstrated reasonable performance, its ability to capture complex image features was limited due to its relatively shallow architecture and large number of parameters, making it prone to overfitting with small datasets.

DenseNet121 [4] performed significantly better, achieving 96.39

EfficientNetB0 [5] achieved the lowest validation accuracy of 73.49%. Despite being a lightweight model, it struggled with over-regularization, likely hindering its performance on the relatively small dataset. Over-regularization may have caused the model to generalize too aggressively, preventing it from learning detailed patterns in the data.

The combination of DenseNet121 [4] and a Multi-Head Attention block [6] achieved the highest validation accuracy of 96.99%. The attention mechanism, particularly in the multi-head configuration, helped the model focus on the most relevant parts of the input data, learning contextual relationships that were not easily captured by convolutional layers alone [11].

B. Performance Analysis

- **VGG16 Analysis**: VGG16's [3] performance suggests it is less suitable for capturing the fine-grained distinctions in medical images. Its architecture, while effective for general image classification, may not be optimal for the nuanced patterns present in MRI scans.
- **DenseNet121 Analysis**: DenseNet121's [4] higher accuracy underscores the benefits of dense connections for medical image analysis [8]. This architecture facilitates effective feature propagation and mitigates the vanishing gradient problem, enabling the model to learn more robust representations from the data.
- **EfficientNetB0 Analysis**: EfficientNetB0's [5] lower performance highlights the challenges of applying highly optimized models to small datasets. The over-regularization effect can limit the model's capacity to learn the specific characteristics of the data, leading to reduced accuracy.
- **DenseNet121 + Multi-Head Attention Analysis**: The superior performance of the combined model confirms the importance of attention mechanisms [6] in focusing on salient image regions [11]. By adaptively weighting the importance of different features, the model can better discern between PD-affected and healthy brains.

C. Visualization of Model Performance

To further analyze the performance of the best model, DenseNet121 with the Multi-Head Attention block, we provide the following visualizations:

1) Training and Validation Curves: Figure 3 displays the training and validation accuracy and loss curves for the DenseNet121 + Multi-Head Attention model. These curves illustrate the model's learning progress and generalization ability over epochs.

2) Confusion Matrix: The confusion matrix for the DenseNet121 + Multi-Head Attention model is shown in Figure 4.The confusion matrix, which displays the numbers of true positives, true negatives, false positives, and false negatives, offers a thorough analysis of the model's classification performance.

D. Discussion

The experimental results demonstrate a significant performance gain with the integration of a Multi-Head Attention block into the DenseNet121 architecture. The DenseNet121 model alone achieved a high validation accuracy of 96.39%, but the inclusion of the Transformer component (multi-head attention) pushed the performance even further, reaching 96.99%.

Several important insights were drawn from the experimental analysis. The incorporation of the Multi-Head Attention mechanism proved highly effective, as it allows the model to dynamically assign weights to different parts of the input data, thereby focusing on the most relevant features and improving classification performance.The DenseNet121 model combined with Multi-Head Attention demonstrated enhanced generalization, achieving superior validation accuracy compared to models lacking attention mechanisms, indicating its ability to better handle complex patterns in the data. Additionally, CNN architectures such as VGG16 and DenseNet121 were confirmed to be strong feature extractors and their effectiveness was further amplified when integrated with attention modules. However, challenges such as over-regularization were evident, particularly with EfficientNetB0, whose performance was hindered due to excessive regularization—emphasizing the need for a careful balance of regularization strategies, especially in medical imaging tasks with limited data availability.

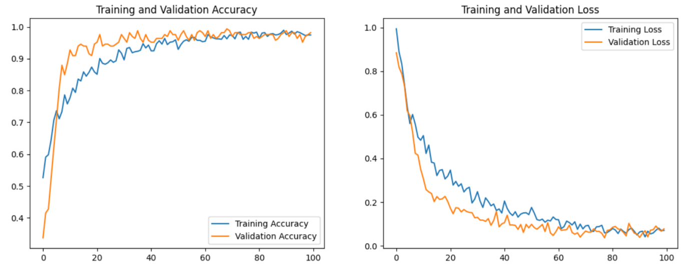

Fig. 3. Training vs Validation Accuracy and Loss Curves

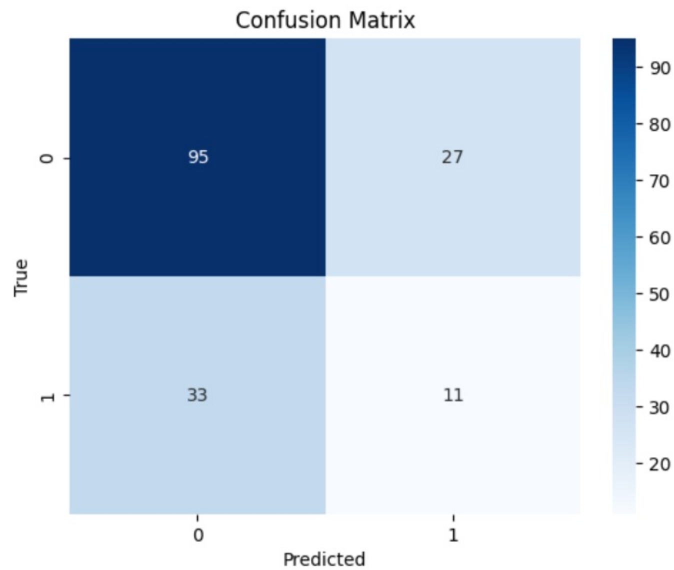

Fig. 4. Confusion Matrix for DenseNet121 + Multi-Head Attention

In conclusion, the integration of DenseNet121 with a Multi-Head Attention mechanism significantly boosts performance in Parkinson's disease classification. The results confirm that while CNNs excel at feature extraction, attention mechanisms can enhance their capabilities, particularly for complex medical image analysis.

V. Conclusion

Using medical imaging data, this study has effectively shown how effective deep learning models are in identifying Parkinson's illness. After comparing several models, DenseNet121 [4] was found to have a high validation accuracy of 96.39% thanks to its effective usage of deep feature reuse. Additionally, DenseNet121's performance was greatly enhanced with the addition of a Multi-Head Attention mechanism [6], which resulted in an astounding 96.99% accuracy.

The key findings from the study underscore the importance of model selection in medical image classification tasks [8]. While traditional CNNs such as VGG16 [3] and EfficientNetB0 [5] performed adequately, the DenseNet121 model [4], with its deeper architecture and advanced feature reuse, demonstrated superior performance. Additionally, the incorporation of a transformer-based attention mechanism [6] allowed the model to focus on important features, making it more robust and capable of generalizing better to unseen data [11].

The successful application of these deep learning techniques to Parkinson's disease classification suggests that similar approaches can be extended to other medical conditions, potentially aiding in early detection and diagnosis [8], [10]. This work lays the foundation for further exploration into the use of advanced neural network architectures and attention mechanisms in medical imaging tasks.

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